Prediction of Poverty Status on Costa Rican Households

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***Abstract*—This project investigates whether supervised machine learning can identify vulnerable households using observable socioeconomic characteristics from the Costa Rican Household Poverty dataset. The original task was formulated as a four-class poverty classification problem, ranging from extreme poverty to non-poor households. After data cleaning, household-level splitting, feature engineering, and feature selection combining correlation, mutual information, and Random Forest importance, several classification models were evaluated, including Logistic Regression, Random Forest, AdaBoost, and KNN. Due to severe class imbalance, techniques such as class weighting and SMOTE oversampling were applied. However, the multiclass models achieved limited performance, with the best macro F1-score reaching only 0.43. Further analysis revealed that poverty levels 1, 2, and 3 strongly overlap in both feature values and distributions, indicating a structural limitation of the dataset rather than a modelling issue. The problem was therefore reframed as a binary classification task distinguishing poor from non-poor households. In this setting, Logistic Regression achieved the best performance, obtaining an F1-score of 0.68 and an AUC of 0.81. The results suggest that observable household features are sufficient to identify households requiring social assistance, but insufficient to reliably distinguish between different levels of poverty severity. Finally, the project discusses practical applications, limitations, and ethical risks related to automated support systems for social aid allocation.**

I. Research Question

How effectively can supervised machine learning models identify the most vulnerable households using observable socioeconomic features, and to what extent can these predictions support more accurate social aid allocation?

II. Problem Definition

Identifying vulnerable households is an important step for governments and NGOs when giving out social assistance. In many countries, welfare targeting depends on slow and expensive manual surveys. This project looks at whether supervised learning can automate poverty classification by using observable household features. This would help make decisions faster. The target variable is a four-level poverty label (1 = extreme poverty, ..., 4 = non-poor), which makes this a multiclass classification task.

More concretely, the model would be used as a prioritising tool, so it can help social workers prioritise which households to inspect or enrol in assistance programmes. The model is not intended to replace human judgment; it's a tool to focus limited resources on the households most likely to be vulnerable.

III. Dataset

The dataset is the Inter-American Development Bank's Costa Rican Household Poverty dataset, published via Kaggle[1]. It contains 9557 individuals across 2988 unique households and 142 features covering housing conditions, asset ownership, education, and household composition.

IV. Data Cleaning and Preparation

A. Target and Household-level split

A normal split would cause household members to appear in both train and test sets and that would cause data leakage, since members of the same household share features and the same label. The split was therefore performed at the household level, all individuals in the same household were assigned to the same target value. An 80/20 stratified split was done to preserve the distribution of classes.

B. Handling Missing Values and Encoding

`v2a1` (rent), `v18q1` (tablets) and `rez_esc` (years of school delayed) were substituted with 0, as NaN logically indicates absence. `Meaneduc` (mean years of education among adults in household) was imputed with the training set mean only, to avoid leakage. Same with `SQBmeaned`, but said mean squared. Since `edjefe` and `edjefa` (education of the household head) only contain the education level of the household head (male or female), both columns were merged into a single numeric feature, and the remaining non-numeric values ("yes" and "no") were imputed using the mode. For the `dependency` variable (dependency rate), "yes" values were replaced with 1, assuming total dependency, while numeric values were divided by 10 to reach a rate.

C. Balance of Classes and Outliers

The target is heavily imbalanced; it is shown that class 4 (non-poor) dominates while class 1 (extreme poverty) is underrepresented. Stratification was applied at split time, and `class_weight='balanced'` was used in all classifiers. No outliers were removed: extreme values such as zero income or severe overcrowding are real signals of extreme poverty, not errors, removing them would harm performance on the most critical class.

D. Feature Engineering

Three features were created to capture more important concepts: `housing_vulnerability` (weighted score of housing type), `asset_count` (total household assets), and `rent_per_person` (monthly rent per household member).

V. Feature Selection

The feature selection process was done to improve the model performance and reduce unnecessary complexity. The objective of it is to keep only the most informative variables.

Once the data was ready, three different methods were applied to conduct feature selection:

First, correlation with the target (`corr`), to measure the linear relationship for each variable and the target. The absolute correlation was used, meaning that the higher its value the stronger the correlation with the poverty level.

Second, mutual information (`MI`), measuring the dependency that each variable had on the poverty level, both for linear and non-linear relationships. This detects important information that correlation might miss.

Third, Random Forest (`RF`), a tree-based model that evaluated the importance of the variables for the model. It detects interactions between complex patterns and the variables.

To get a solid selection and avoid depending on an only criterion, a combination of the three methods was done. By assigning a rank to each of the variables in each model, an average ranking was done.

When evaluating the ranking, a redundancy in variables was identified. There were variables that were the squared version of others (those starting with `SQB`). The one having the best ranking was kept as both represented the same information.

Finally, to choose the most favourable number of features, Logistic Regression was used as a baseline model. The performance was evaluated with different numbers of features, being the one with the highest accuracy selected as optimal number of features.

The seventeen features selected in this process will be the ones used in the modelling phase.

VI. Model Implementation and Evaluation

A. Multiclass Classification

The original task was intended to be a four-class classification (extreme poverty, moderate poverty, vulnerable, non-poor). To address the severe class imbalance, we tested two strategies across four algorithms (Logistic Regression, Random Forest, AdaBoost, KNN):

- Modified class weighting in grid search.
- Synthetic oversampling (SMOTE).

Neither produced meaningful results. The best model was Logistic Regression with SMOTE, with just a 0.43 F1-macro score, with all other models scoring notably worse. These results indicate the problem is not sampling imbalance alone.

B. Binary Classifications Results

All four models were re-run on a binary version of the problem and shows a clear improvement than the multiclass version and reframing the problem improved the performance. Considering the F1 on the majority class, the models ranked in the following order: Logistic Regression, Random Forest, AdaBoost and KNN. Logistic Regression achieved the best F1 of 0.68 with an accuracy of 0.74 and recall of 0.75, meaning that it correctly classifies 75% of genuinely poor households, while out of all the households flagged as poor, 62% are. However, Random Forest followed closely with an F1 slightly under of 0.65 but a higher precision meaning that it's more likely to be correct when flagging a household as poor at the cost of more false negatives. The non-poor class was predicted with greater confidence across all models, which reflects that non-poor households have more distinctly separable features. The ROC curve of Logistic Regression yields an AUC of 0.81 meaning that the model correctly ranks a randomly chosen poor household above a randomly chosen non-poor one 81% of the time.

C. Structural Limitation Analysis

To understand whether the poor performance was a modelling limitation or a data limitation, we investigated whether the features themselves were distinguishable enough between poverty levels 1, 2 and 3. We computed feature means and standard deviations by class and then calculated all six pairwise absolute differences between class means and standard deviations, normalizing each feature's differences to a 0-1 scale so that features with different units could be compared directly (0 meaning identical distribution, 1 meaning maximum separation).

The results revealed a structural problem, since classes 1, 2 and 3 are very similar in values and in distribution. Mean education, for example, ranges only from 6.7 to 7.8 years across the three poverty levels, in contrary of the 10.3 years for class 4. The pairwise scores confirmed this pattern across all features, as separations between poor classes consistently fell between 0.01 and 0.35 (with few exceptions), while any comparison involving class 4 scored 0.6 to 1 (with few exceptions). To confirm this was not simply a difference in means (but a deeper distributional overlap), we also analysed the pairwise differences (A1) in standard deviations across classes. The pattern mirrors the means analysis. Variance difference between poor classes is consistently low, mostly between 0.02 and 0.5, while comparison involving class 4 score notably higher across nearly all features. This indicates that poor households not only share similar average values but also similar spread, meaning their distributions genuinely overlap rather than simply having close centres.

To rule out that this is due the feature selection being focused on class 4, we repeated the analysis restricting feature selection to poor households only, giving the model every possible advantage. The results remained similar. The most difference we could find was between classes 1 and 3, which makes sense since they are in opposites sides of the poverty spectrum. However, due to class 2 not having distinguishable values from either neighbour, it continues to blur the boundaries across all three poverty levels, leaving overall separation power significantly low even with an optimized feature set.

Additionally, we analysed the error structure of the best multiclass model. We found that 64.1% of its mistakes were adjacent-class confusions, meaning the model consistently placed households one level off on the poverty scale rather

than randomly guessing. This suggests that the model can identify poverty as a spectrum, but the available features do not have enough clear boundaries between severity levels to carry a four-class prediction. No standard supervised approach can resolve this data limitation.

D. Final model

Given these findings, the problem was reframed as binary classification, non-poor (class 4) and poor (classes 1-3), with the poor as the positive class. This way we could achieve operational value supported by the data. All four algorithms were re-evaluated, tuned with a 5-fold stratified cross-validation grid search. F1 on the poor class was used as the scoring metric to ensure that the model's ability to correctly identify poor households, the actionable outcome, drove the selection rather than overall accuracy. Logistic Regression achieved the best performance, although the rest of the models performed closely behind, supporting the conclusion that the binary boundary is genuinely learnable from the available features.

VII. Critical Reflection

The main achievement of this project was understanding why the multiclass model failed and identifying an alternative model that produced actionable results. The observable characteristics found in the households (education, quality of life, assets, etc.), cannot distinguish between levels of poverty within the poor population. Families in different levels of poverty often live in materially similar conditions. The difference between poverty levels 1 and 2, may differ in some specific types of features like income flows or health treatments, which it does not appear in this dataset. This finding has direct practical value, as the results suggest that the right question to ask of this type of data is binary: does this household need attention, or not? That question is answerable, actionable, and as our binary model shows learnable with reasonable performance. If a government genuinely needs to distinguish poverty severity at scale, it will require a more extensive dataset with more other features, and that would require a more complex and difficult recollection of data.

Concretely, the binary model can support two types of decisions. First, it can flag households predicted as poor that are currently receiving no assistance. In these cases, a social worker would conduct a field inspection to investigate why the household is excluded from the system. For example, because they are not registered in the welfare programme, lack the required documentation, or are unaware of their eligibility. This would allow governments to actively close registration gaps and reduce exclusion errors.

Second, it can identify households predicted as non-poor that are currently receiving aid. Here, a case review would investigate whether the assistance is still justified. For example, whether the household's economic situation has improved since enrolment. This would help reallocate resources towards households that genuinely need them, while also maintaining the integrity of the programme.

Finally, the model can help prioritise where to deploy outreach teams in areas with a high concentration of predicted poor households but not yet registered in any assistance programme. Rather than spreading resources uniformly, governments or NGOs could focus mobile registration units or awareness campaigns in those communities first. Importantly, the model would guide where to look, not who receives aid. The final decision would still be made at the individual household level.

VIII. Risks

A. Data Risks

The model may suffer from distributional shift, as it is trained on 2016 data that does not reflect current socioeconomic conditions. In addition, we find sampling bias, where the dataset may not represent the whole population and some living conditions may be underrepresented such as rural zones.

B. Modelling Risks

One of the main risks of modelling in our case is the oversimplification of a complex social reality, leafing to what is known as *construct gap*. The model relies only on a limited set of observable household features, which are unlikely to fully capture the multidimensional nature of poverty. As a result, it may struggle to distinguish between similar levels, such as extreme poverty (level 1) and moderate poverty (level 3), since these differences are not always reflected in a few measurable variables. This limitation can lead to reduced predictive performance and highlights a fundamental constraint of the approach.

C. Evaluation Risks

Class imbalance represents a fundamental limitation in our problem: even when using metrics such as macro F1, the minority classes remain underrepresented and harder to learn. As a result, the models have performed poorly on these groups, and although several mitigation techniques were explored, none of them led to a substantial improvement. Due to this risk, we chose to use a binary model, which mitigates the negative effect of the class imbalance but does not eliminate it completely.

D. Ethic and Social Risks

From an ethical and social perspective, the model's errors can have real consequences. Looking at the confusion matrix (A2), some households in poverty are classified as non-poor and vice versa (e.g., 295 and 188 cases), which means they could be left without the support they actually need or a possible misallocation of resources. This shows that the model's mistakes are not just technical, but can directly affect how aid is distributed in practice. Additionally, if the model is used to support human decision-making, there is a risk of automation bias, where users may rely too heavily on the model's predictions even when they are incorrect and these errors could be amplified.

REFERENCES

[1] Kaggle, "Costa Rican Household Poverty Prediction," 2018. [Online]. Available: https://www.kaggle.com/competitions/costa-rican-household-poverty-prediction/overview

APPENDIX

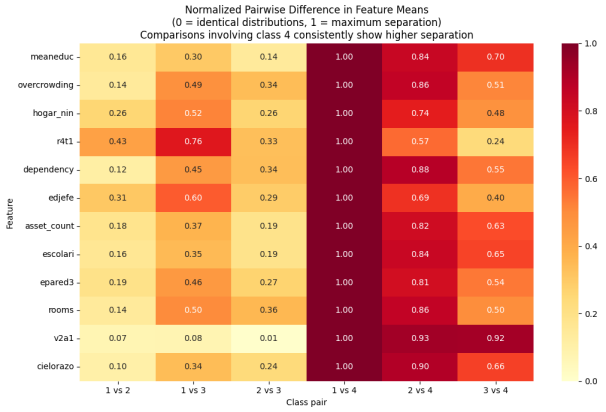


Fig. A1. Pairwise model performance comparison.

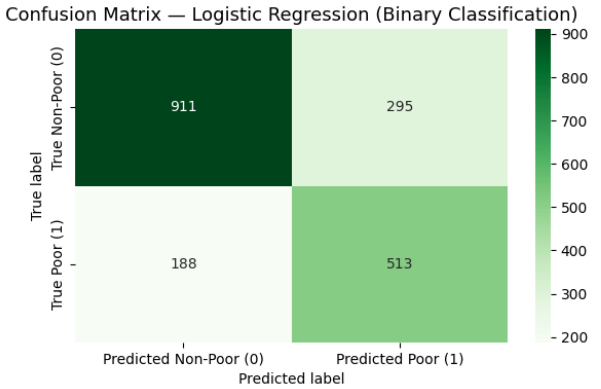


Fig. A2. Confusion matrix of the binary classifier.